

Date: Feb 1, 2021

Experiment: 4

Title: Survey of some functional groups

Objective

The objective of this experiment was to study the characteristics chemical properties of some functional groups

Theory

A functional group: is a substituent or moiety in a molecule that causes the molecule's characteristic chemical reactions. The same functional group will undergo the same or similar chemical reactions regardless of the rest of the molecule's composition. In our experiments alkanes, alkenes, alkynes, aromatics were used

Alkane: are not very reactive when compared with other chemical species. are also less dense than water, as one can observe, oil, an alkane, floats on water. are non-polar solvents. Since only C and H atoms are present, alkanes are nonpolar. Alkanes are immiscible in water but freely miscible in other non-polar solvents. The general formula for alkanes is C_NH_{2N+2} .

Alkene: are also insoluble in water, are also less dense than water. They contain a carbon double bond.

Alkynes

Generation of acetylene:

Acetylene is the simplest alkyne group with a carbon number 2. It can be prepared in the laboratory using the following reaction:



(Calcium carbide) + water $\longrightarrow$ acetylene + calcium hydroxide

Bromination test: is a qualitative test for the presence of unsaturation (carbon-to-carbon double or triple bonds), phenols and anilines. An unknown sample is treated with a small amount of elemental bromine in an organic solvent, being as dichloromethane or carbon tetrachloride. Presence of unsaturation and/or phenol or aniline in the sample is shown by disappearance of the deep brown coloration of bromine when it has reacted with the unknown sample.

Baeyer's test: for unsaturation is for determining the presence of carbon-carbon double bonded compounds, called alkenes or carbon-carbon triple bonded compounds, called alkyne bonds. The

Baeyer test uses dilute Potassium Permanganate to oxidize the carbon-carbon double or triple bond.

Apparatus used

- Test tubes
- Delivery tube
- Clamp
- Distillation flask
- Boss head
- Stand
-

Chemicals used

- Sulphuric acid(H_2SO_4)
- Petroleum ether (C_6H_{14})
- Ethanol($\text{C}_2\text{H}_5\text{OH}$)
- Water
- N- hexane
- Kerosene
- Bromine in carbon tetrachloride (5%)
- Potassium permanganate (KMnO_4)
- Cyclohexene
- Benzene/ toluene
- Acetone
- Phenol
- Ice
- Nitric acid
- DNPH
- Neutral ferric chloride
- Calcium carbide(CaC_2)

Procedure

A. Alkanes: solubility test

N-hexane

- Sulphuric acid, petroleum ether, ethanol, water, N-hexane were set on a table as a solvent for the experiments by our instructor. Dry test tubes were prepared and were put on a test tube holder. 5 drops of N-hexane was added in 4 dry test tubes. Following that, 3ml of

sulphuric acid, 3 ml of petroleum ether, 3 ml of ethanol, and 3 ml of water were added to 4 different dry test tubes. 3 ml of the sulphuric acid was added into one of the test tubes containing 5 drops of N-hexane, 3 ml of petroleum ether was added to the other test tube containing 5 drops of N-hexane, 3ml of ethanol was added to the third test tube containing 5drops of N-hexane, and lastly 3ml of water was added to the 4th test tube containing 5 drops of N-hexane. The observed results were recorded in the form of a table. After completing the above steps, we properly poured the mixture, and washed the test tubes. The test tubes were washed again by ethanol so it could dry them properly.

Kerosene

- Sulphuric acid, petroleum ether, ethanol, water, Kerosene were set on a as a solvent on a table for the experiments by our instructor. Dry test tubes were prepared and were put on a test tube holder. 5 drops of kerosene were added in 4 dry test tubes. Following that, 3ml of sulphuric acid, 3 ml of petroleum ether, 3 ml of ethanol, and 3 ml of water were added to 4 different dry test tubes. 3 ml of the sulphuric acid was added into one of the test tubes containing 5 drops of kerosene, 3 ml of petroleum ether was added to the other test tube containing 5 drops of kerosene, 3ml of ethanol was added to the third test tube containing 5drops of kerosene, and lastly 3ml of water was added to the 4th test tube containing 5 drops of kerosene. The results were observed and the recorded in the form of a table. After completing the above steps, we properly poured the mixture, and washed the test tubes. The test tubes were washed again by ethanol so it could dry them properly. After completing the above steps, we properly poured the mixture, and washed the test tubes. The test tubes were washed again by ethanol so it could dry them properly.

Alkane: Reaction of N-hexane

- N-hexane, petroleum ether, kerosene, a solution of bromine in carbon tetrachloride (5%) were set on a table for the experiments by our instructor. Dry test tubes were prepared and were put on a test tube holder. 4-5 drops of a solution of bromine in carbon tetrachloride (5%) were added in 3 dry test tubes. 1ml portion of petroleum ether, 1ml of N-hexane, 1ml of kerosene was added in 3 different dry test tubes. 5 drops of a solution of bromine in carbon tetrachloride (5%) were added to the first test tube containing 1ml of petroleum ether, 5 drops of a solution of bromine in carbon tetrachloride (5%) were added to the second test tube containing 1ml of N-hexane, 5 drops of a solution of bromine in carbon tetrachloride (5%) were added to the third test tube containing 1ml of kerosene. After completing the above steps, all the test tubes were put it the dark for 15 minutes in a cupboard that serves as storage. 15minutes later, to test the presence of HBr a litmus paper was used. Next, the above steps were repeated, but the second time all the test tubes were put to next to the window for a light source. Again to test the presence of HBr a litmus paper was used. The observed results were recorded in the form of a table. After completing the above steps, we properly poured the mixture, and washed the test

tubes. The test tubes were washed again by ethanol so it could dry them properly. The observed results were recorded in the form of a table.

B. Alkene: solubility test

- Cyclohexene and sulphuric acid were set on a table for the experiments by our instructor. Dry test tubes were prepared and were put on a test tube holder. 10 drops of cyclohexane was added to a test tube. 3 drops of the solvent sulphuric acid was added to another test tube. The cyclohexene was added to the test tube containing the sulphuric acid. After completing the above steps, we properly poured the mixture, and washed the test tubes. The observed results were recorded in the form of a table. The test tubes were washed again by ethanol so it could dry them properly.

Alkene: reaction with bromine in carbon tetrachloride test

- Cyclohexene and a solution of bromine in carbon tetrachloride (5%) were set on a table for the experiments by our instructor. Dry test tubes were prepared and were put on a test tube holder. 0.5ml (10 drops) of cyclohexene was added to a test tube. 3 drops of bromine in carbon tetra chloride was added in another test tube, and then it was added to the test tube containing the cyclohexene. The observed results were recorded in the form of a table. After completing the above steps, we properly poured the mixture, and washed the test tubes. The test tubes were washed again by ethanol so it could dry them properly

Alkene: reaction with aqueous potassium permanganate

- Cyclohexene and potassium were set on a table for the experiments by our instructor. Dry test tubes were prepared and were put on a test tube holder. 0.5ml of cyclohexene was added to a test tube. 1ml of 0.5% aq. Potassium permanganate was added to another test tube, and it was added to the test tube containing the cyclohexene. The observed results were recorded in the form of a table. After completing the above steps, we properly poured the mixture, and washed the test tubes. The test tubes were washed again by ethanol so it could dry them properly.

Alkynes

- The apparatus was set by our instructors, as they were the once who performed this reaction. Calcium carbide was put inside the distillation flask. At the top water was added slowly in the form of a droplet, and when the water reached the distillation flask, it reacted with calcium carbide which produced acetylene in the form of gas. The acetylene will pass through the delivery tube. According to the first test, acetylene was generated, to check that, our instructor lit a match and put it closer to the delivery tube. Then a flame was released. The second test was bromination test. Bromine was reacted with acetylene, and the result was observed. The third test was also a reaction. According to Baeyer's test, potassium permanganate reacted with acetylene, and the result was observed and recorded.

C. Aromatic hydrocarbons: test for saturation

- Benzene and a solution of bromine in carbon tetrachloride were set on a table for the experiments by our instructor. Dry test tubes were prepared and were put on a test tube holder. 6 drops of benzene was added to a test tube. 3 drops of a solution of bromine in carbon tetrachloride was added to another test tube, and it was added to the test tube containing the benzene. After completing the above steps, the reaction of benzene with a solution of bromine in carbon tetrachloride was observed, and the results were recorded. After completing the above steps, we properly poured the mixture, and washed the test tubes. The test tubes were washed again by ethanol so it could dry them properly.

Aromatic: nitration of toluene

- Toluene, concentrated sulphuric acid, nitric acid were set on a table for the experiments by our instructor. Dry test tubes were prepared and were put on a test tube holder. 10 drops of concentrated sulphuric acid was added to a test tube following that 3 drops of concentrated nitric acid was added to the test tube. Then 4 drops of test tube was added to the test tube containing the concentrated sulphuric acid and the concentrated nitric acid. The test tube was shaken, and left to stand for 15 minutes. Pieces of ice were added to a larger test tube. After 15 minutes, the mixture was poured into the larger test tube containing ice. After completing the above steps, the nitration of toluene was observed. The results were recorded. After completing the above steps, we properly poured the mixture, and washed the test tubes. The test tubes were washed again by ethanol so it could dry them properly.

Ketone: Test of acetone

- Acetone and DNPH were set on a table for the experiments by our instructor. Dry test tubes were prepared and were put on a test tube holder. 10 drops of acetone was added to a test tube. Few drops of DNPH were added to another test tube, and it was added to the test tube containing acetone until a precipitate was observed and the results were recorded. After completing the above steps, we properly poured the mixture, and washed the test tubes. The test tubes were washed again by ethanol so it could dry them properly.

Phenol: test of phenol

- Phenol and 2% of neutral ferric chloride were set on a table for the experiments by our instructor. Dry test tubes were prepared and were put on a test tube holder. 20 drops of 10% aq. solution of phenol was added to a test tube. 3 drops of 2% of neutral ferric chloride solution was added to another test tube, and it was mixed with the test tube containing the phenol. The observed results were recorded in the form of a table.

After completing the above steps, we properly poured the mixture, and washed the test tubes. The test tubes were washed again by ethanol so it could dry them properly.

Result and discussion

A. Alkane: solubility test

	solvent			
Solute	Sulphuric acid(H₂SO₄)	Petroleum ether	Ethanol	Water
N-hexane	Insoluble	soluble	soluble	Insoluble
Kerosene	insoluble	soluble	soluble	insoluble

- N-hexane is a non-polar hydrocarbon, and according to the like dissolves like principle, N-hexane should be soluble in non-polar solvents. The solvents that we used are sulphuric acid which is polar, petroleum ether which is non-polar, ethanol which is polar, and water which is also polar. Accordingly, N-hexane should have been soluble in only petroleum ether because it is the only non-polar solvent that was used in our experiment. However N-hexane was soluble in ethanol, and after asking our instructor how that was possible, we have come to the understanding that the amount of Ethanol sample that we were told to use was small, and that was why the N-hexane became soluble in the Ethanol. Increasing the amount of Ethanol sample will result in the insolubility of N-hexane solute. From this experiment we can conclude that alkanes are only soluble in non-polar solvents like gasoline and petroleum ether.
- When we look at kerosene we observe the same results as N-hexane, and this is because kerosene is non-polar just like N-hexane

Alkane: reaction of N-hexane in the dark

	N-hexane	Petroleum ether	kerosene
Bromine(CCl ₄)	No reaction(No color change)	No reaction(no color change)	No reaction(no color change)

Alkane: reaction test of N-hexane in the light

	N-hexane	Petroleum ether	kerosene
Bromine(CCl ₄)	Reaction(from blue to orange/yellow)	Reaction(from blue to orange/yellow)	Reaction(from blue to orange/yellow)

- Bromine has a reddish brown color, after the addition of N-hexane, petroleum ether and kerosene in different test tubes, and after putting it in a dark area, the color of bromine doesn't change it still remains the same. This is because the purpose of the light source especially UV ray is to split the bromine molecule into two reactive radicals, resulting in a very slow loss of amber bromine color. The radical bromine atoms can then go on to react with other species in solution (including each other) in a chain reaction. So the absence of light is the cause for the above result.

B. Alkene: solubility test

	Cyclohexene
Conc. H ₂ SO ₄	Immiscible

- We have observed that cyclohexene is not soluble in sulphuric acid, the cause for this is the difference in polarity. Cyclohexene is a non-polar hydrocarbon while sulphuric acid is polar so they can't dissolve in each other, instead they form a layer.

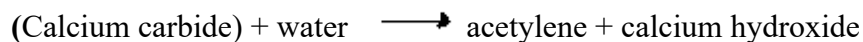
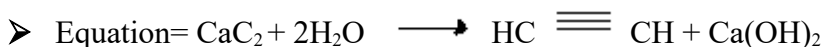
Alkene: reaction test

	Cyclohexene
Bromine(CCl ₄)	Reaction(from reddish brown to colorless)
Potassium permanganate(KMnO ₄)	Reaction(from pink/purple to brown color change)

- Cyclohexene reacts with both bromine solution in carbon tetrachloride the double bond breaks, and a bromine atom becomes attached to each carbon. The bromine loses its original red-brown color to give a colorless liquid, and in the case of potassium permanganate cis-1,2-cyclohexanediol.
- Equation=

C. Alkyne

- Reaction of calcium carbide with water results in the formation of acetylene.



- Bromination test is a test which is useful for checking whether a hydrocarbon is saturated or unsaturated. Alkynes have triple bonds which means there will be a color change observed when they react with bromine. **This color change was from the reddish brown color to colorless.**
- **Potassium permanganate was reacts with alkynes, and in this case the pink/purple color of potassium permanganate changed to brown color.**

D. Aromatic: test for unsaturation

	Bromine (CCl ₄)
Benzene	No reaction(no color change)

Benzene molecule is unsaturated but it does not undergo electrophilic addition reaction because saturating the carbon-carbon bonds destroys the delocalized pi-cloud and as we know that this delocalized pi-structure is very stable. So due to this benzene does not decolourise bromine water. That is why the color of bromine remained unchanged.

	H ₂ SO ₄ and HNO ₃
Toluene	Reaction(from colorless to yellow)

Toluene underwent a reaction and after the addition of ice a yellow color was observed.

➤ Equation=

	DNPH
Acetone	Forms precipitate

The presence of a precipitate is an indication that a ketone is in the solution.

➤ Equation=

Quiz question

1. Give the mechanisms of the reaction of DNPH with acetone?

➤ Equation=

2. What is the difference between addition and substitution reactions?

- An addition reaction occurs when two or more reactants combine to form a single product. This product will contain *all* the atoms that were present in the reactants.

Addition reactions occur with unsaturated compounds. The general equation for an addition reaction: $A+B \rightarrow C$

- Substitution reaction occurs when an exchange of elements in the reactants takes place.

The initial reactants are transformed or *swopped around* to give a final product. The general equation for a substitution reaction: $AB+CD \rightarrow AD+BC$

3. Give mechanisms for the nitration of toluene?



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DATE OF SUBMISSION: FEB 8, 2021